

Aatika Fatima

SENIOR JAVA BACKEND ENGINEER

AI-Assisted Design and Development | Claude by Anthropic | Spring AI | Accelerated Delivery and Code Quality
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PROFESSIONAL SUMMARY

Senior Software Engineer with 12+ years of experience specialising in B2B payment platforms, virtual card lifecycle management, and enterprise microservices. Deep expertise integrating global payment providers, designing intelligent card-selection engines, and building fraud-prevention systems at scale. Proven ability to own solutions end-to-end - from architecture and development through deployment and production operations - reducing manual workload and maximising payment profitability. Actively expanding into applied AI engineering, incorporating Spring AI and Claude-based tooling into the Java/Spring ecosystem to accelerate delivery and code quality.

SKILLS & COMPETENCIES

- B2B Credit Payments
- Card Selection Optimisation
- RESTful & Financial APIs
- Generative AI & Agentic Systems
- Multi-Bank API Integration
- Authorization Systems Design
- Payment Architecture
- Retrieval-Augmented Generation (RAG)
- Fraud Detection & Prevention
- Chargeback & Rebate Strategy
- Prompt Engineering
- Model Context Protocol (MCP)

TECHNICAL SKILLS

Languages: Java, Kotlin, SQL

Frameworks: Spring Boot, Spring MVC, Spring Cloud, Spring Data JPA, Spring Security, Spring AI, Hibernate, JPA

Architecture: Microservices, REST APIs, Distributed Systems, Event-Driven Architecture, Enterprise Design Patterns

Payments & FinTech: B2B Payment Platforms, Virtual Card Lifecycle, Fraud Detection, Card Authorisation, PCI Security, Payment Optimisation

Payment Providers: WEX, VNett, Citi, Revolut, Amadeus, Adyen, AMEX, Cornercard

Databases: Oracle, SQL, Azure SQL

Messaging: RabbitMQ

Monitoring: Graylog, Grafana, New Relic

Testing: JUnit, Spring Test, Unit Testing, Integration Testing

AI / GenAI Tooling: Spring AI, Generative AI, Large Language Models (LLM), Claude API, Claude Code, Agentic AI, AI Agents, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Database, Semantic Search, Embeddings, Model Context Protocol (MCP), Tool Calling / Function Calling, Conversational AI, Natural Language Processing (NLP), AI-Assisted Development, LLM-Enabled Java Applications

Cloud: Azure, AWS

Practices: Agile, CI/CD, End-to-End Ownership, AI-Assisted Development (Claude Code, Spring AI)

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Payments Platform

Dec 2021 - Jun 2026

Lastminute.com | Hyderabad, India

Tech Stack: Java, Kotlin, Spring Boot, Spring Data JPA, Spring MVC, Spring Cloud, Spring AI, Generative AI, LLM, Agentic AI, RAG, MCP, Microservices, REST APIs, Distributed Systems, Payment Systems, Virtual Cards, Fraud Detection

Payment Platform Architecture & Integrations

- Designed and built a distributed B2B payment platform processing 10,000+ travel transactions monthly across flights, hotels, baggage, parking, and ancillary services - owning the full lifecycle from architecture to production deployment.
- Integrated 6 global payment providers - WEX, VNett, Citi, Revolut, Amadeus, and Adyen - achieving 99.9% payment availability through multi-provider redundancy.
- Negotiated and implemented Adyen integration with a 98% rebate rate, directly improving payment profitability.

- Designed a Payment Decision Engine that automatically selects the most profitable card from a pool of 8 providers based on FX rates, rebates, spending limits, and availability - eliminating 100% of manual card selection.
- Built failover logic into the Payment Decision Engine to automatically serve the next best card on failure, reducing manual booking intervention by 95% and saving ~200 hours/month in operations effort.
- Implemented low-balance detection to auto-suspend card issuance from depleted banks, preventing cascading failures and reducing payment decline rates by ~40%.

Virtual Card Lifecycle & Security

- Architected and delivered microservices for the complete virtual card lifecycle: Card Issuance Service, Card Modification Service, Card Fetch Service, and Card Cancellation Service.
- Built an automated Card Cancellation Scheduler that cancels virtual cards within 2 hours of booking confirmation, preventing unauthorised charges and eliminating card misuse across 100% of issued cards.
- Enforced single-use virtual card policy for WEX and Adyen, reducing exposure to repeated or unauthorised transactions by 100%.
- Encrypted all sensitive card data (PAN, CVV, expiry) at rest and in transit, achieving full PCI DSS compliance across 100% of virtual card transactions.
- Developed WEX authorisation workflows to detect and block fraudulent transaction attempts in real time, reducing fraud-related chargebacks by ~30%.

Finance Operations & Observability

- Built a self-serve Finance Team UI for virtual card management, eliminating engineering dependency for card operations and saving ~15 hours/week of developer time.
- Supported manual booking operations by issuing credit cards on demand, enabling the finance team to independently fulfil 100+ manual bookings per month without engineering support.
- Developed a bulk resubmit mechanism for failed manual bookings during production incidents, reducing incident recovery time by ~70% and enabling batch reprocessing of 500+ bookings without developer intervention.
- Developed a real-time Account Balance API surfacing live balances across 8 bank accounts in the finance dashboard, enabling proactive spend monitoring.
- Implemented Graylog and Grafana observability across 5+ payment microservices, reducing mean time to detect (MTTD) production issues by ~60%.
- Migrated a legacy Spring MVC monolith to Spring Boot microservices, cutting new bank integration time from 3-4 months to ~1 month - a 75% reduction in time-to-market.

AI-Powered Flight Search Assistant (searchFlights) - Generative AI & Agentic Systems

- Designed multi-step AI agents using Spring AI, Kotlin, and Large Language Models (LLMs) for the searchFlights module, capable of collecting missing trip details (origin, destination, dates, passenger count), validating search requests, ranking flight results by price, duration, and rebate value, and orchestrating the end-to-end flight search workflow - a core Agentic AI / AI Agent implementation.
- Applied prompt engineering to build structured prompt templates and JSON-schema response frameworks, enabling reliable, machine-readable LLM outputs for downstream booking services.
- Implemented Relevancy Evaluation and response-scoring workflows to measure and improve AI-ranked flight result quality, ranking accuracy, and retrieval precision.
- Designed Tool Calling / Function Calling architectures enabling AI agents to autonomously invoke external services during flight search - fare/pricing APIs, availability checks, payment systems, and internal booking platforms.
- Built Conversational AI workflows supporting multi-turn flight search interactions (e.g., refining dates, swapping airports, adjusting passenger mix) with context retention and context-aware decision making.

Retrieval-Augmented Generation (RAG) & Semantic Search

- Designed RAG (Retrieval-Augmented Generation) pipelines using vector databases, embeddings, and semantic search to ground LLM responses in factual flight policy, fare-rule, and baggage/ancillary data, reducing hallucination.
- Evaluated document-ranking and information-retrieval strategies to support customer-support automation and self-service query resolution around flight search and booking questions.

Model Context Protocol (MCP) & Tool Integration

- Designed MCP (Model Context Protocol) architectures enabling secure, structured communication between the searchFlights AI agent and enterprise services (fare, availability, and booking systems).

- Defined integration patterns connecting LLMs to external flight-search tools, provider APIs, and business systems through Spring AI.

AI-Assisted Development Powered by Claude

- Leveraged Claude Code across the full SDLC - code generation, refactoring, debugging, and test creation - reducing average feature delivery time by ~30%.
- Used Claude to analyse Graylog logs and identify root-cause bugs in production, cutting mean time to resolution (MTTR) by ~50% on critical payment incidents.
- Applied AI-assisted development to accelerate delivery of WEX authorisation workflows, with Claude generating boilerplate, suggesting edge-case handling, and reviewing logic for correctness.
- Integrated Claude into the full development lifecycle: local testing, SonarQube code coverage checks, and pre-commit review - ensuring higher code quality with faster turnaround.

Software Developer

Oct 2020 - Oct 2021

HCL Technologies (Client: Gap IT Services) | Hyderabad, India

Tech Stack: Java, Spring Boot, Spring Microservices, JPA, Hibernate, Oracle, REST APIs, RabbitMQ, JUnit

Project: Receipt Visibility System - microservices-based reporting platform consolidating data from multiple downstream systems to generate business intelligence reports.

- Built 5+ Spring Boot microservices and REST APIs powering a BI reporting platform consolidating data from 10+ downstream systems.
- Implemented JPA/Hibernate persistence layers and applied validation using JSR-303 and Hibernate Validator.
- Implemented RabbitMQ listener retry mechanisms, achieving a 99%+ message processing success rate for async reporting pipelines.
- Configured New Relic monitoring, production alerting, and resolved security vulnerabilities.

Software Developer

Jan 2016 - Oct 2020

Accenture Ltd | Hyderabad, India

Tech Stack: Java, Spring, Spring MVC, Spring Security, Hibernate, JPA, SQL, J2EE

- Developed enterprise Java applications using Spring, Hibernate, and J2EE best practices across multi-year client engagements.
- Implemented authentication and authorisation using Spring Security; built structured exception-handling frameworks.
- Designed database schemas, optimised SQL queries, and built Hibernate DAO/persistence layers.
- Applied design patterns including MVC, DAO, Singleton, Factory, and Session Facade for scalable, maintainable architecture.
- AWS EC2: Experience in provisioning and managing EC2 instances for Spring Boot application deployment, including server configuration, security groups, networking, and application monitoring.
- AWS S3: Experience in using S3 for object/file storage, file upload/download, access permissions, and application integration.
- AWS IAM: Experience in managing IAM users, roles, and policies to control secure access to AWS resources and services.

Senior Java Instructor

Jun 2011 - Jan 2016

NIIT Ltd | Hyderabad, India

Delivered advanced Java and enterprise software training; developed curriculum and mentored students in Spring, Hibernate, and software engineering best practices.

Computer Science Lecturer

Jun 2009 - Jun 2011

Dr. VRK Women's College of Engineering and Technology | Hyderabad, India

Lectured on core computer science and programming fundamentals for undergraduate engineering students.

EDUCATION

Bachelor of Computer Science & Engineering

Sep 2004 - May 2008

Osmania University, Hyderabad, India

CERTIFICATION

AWS Certified Developer Associate